MASAAI MARA UNIVERSITY

REGULAR UNIVERSITY EXAMINATIONS
2022/2023 ACADEMIC YEAR
SECOND YEAR SECOND SEMESTER

SCHOOL OF PURE APPLIED AND HEALTH SCIENCES
BACHELOR OF SCIENCE
IN COMPUTER SCIENCE

COURSE CODE: COM 2221-1
COURSE TITLE: SOFTWARE ENGINEERING

DATE: 18/4/2023 **TIME:1100-1300 HRS**

- Question ONE in section A is COMPULSORY.
- Answer any two [2]questions from section B
- Illustrate your answers were necessary
- SWITCH OFF your mobile phones.
- Any form of cheating will be penalized.

SECTION A

QUESTION ONE

a) You have been asked to develop the following software project:

Masaai Mara University Cafeteria Enquiries and Orders Information Management System. The client wants development of a new management information system (MIS) to enable multi-user data entry and querying. It is expected that the system will encompass the following areas:

- User friendly interface
- Data entry validation
- Defined reports, especially job control sheets, which the client already uses on paper
- Querying of data to produce reports.

The client has a SQL server system that should be the database and will support intranet access from several desks. The development models suggested for this project are: incremental development using a series of prototypes and waterfall model with progression through analysis, design, coding, testing and maintenance. Determine the suitability of each of these two life cycles models for this project. Use appropriate examples to support your answers. **[8 Marks]**

b) Outline FOUR qualities of a good system analyst **[2Marks]**

c) Differentiate between Software Engineering **methods**, **tools** and **procedures.** **[3Marks]**

d) Explain FOUR benefits that an incremental software development process model might have compared to the waterfall model **[4Marks]**

e) It has been argued that "software reuse is the only realistic approach to bring about the gains of productivity and quality that the software industry needs". Comment on this statement, making arguments for and against its validity. J **[3Marks]**

SECTION B

QUESTION TWO

a) Describe the various different types of non-functional requirements which may be paced on a system. Give examples of these requirements. **[4Marks]**

b) Exam Management System (EMS) is the all in-one solution for managing exams at your institution. Optimize your students' experience by reducing the operational risks which exams produce, and save time and money in the process. Write a set of non-functional requirements for an examination management system **[6Marks]**

c) Suggest how a software engineer responsible for drawing up a system requirements specification might keep track of the relationships between functional and non-functional requirements of a system. **[5Marks]**

QUESTION THREE

a) Explain the following prototyping approaches in software process

i. Evolutionary prototyping **[2Marks]**

ii. throw-away prototyping **[2Marks]**

b) The four categories of software maintenance are: Perfective, Adaptive, Corrective and Preventive. Explain how would you classify the following maintenance activities:

(i) hardware and software platform change, **[1Mark]**

(ii) correcting errors found by users, **[1Mark]**

(iii) producing a design document (as the original document has been lost) **[2Marks]**

(iv) Modifying some parts of software due to changing user requirements. **[2Marks]**

c) Write test cases for the login page for a social networking service (e.g. Twitter, Facebook, etc.). Include both positive and negative test cases. **[5Marks]**

QUESTION FOUR

a) with an aid of a diagram, describe **level o** and **level 1** on DFD **[5Marks]**

b) Explain the four components of DFD **[2Marks]**

c) Differentiate between **black-box** and **white-box testing** **[4Marks]**

d) Write short notes on the following as used in software design

i. cohesion **[1Mark]**

ii. coupling **[1Mark]**

iii. modularization **[1Mark]**

iv. Concurrency **[1Mark]**

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